

Hungi Jung

junghungi@kakao.com caribbeanbay.github.io/hungi-jung.github.io
Google Scholar ORCID OpenReview LinkedIn

Medical AI researcher with experience across clinical text, longitudinal EHR data, physiological signals, and medical imaging. My research focuses on developing multimodal learning approaches that integrate heterogeneous clinical data for patient-level prediction and clinically meaningful decision support.

Education

B.S. in Computer Science, Soongsil University *Mar 2022 – Feb 2025*
GPA: 3.26/4.5 (Major GPA: 3.51/4.5)

Experience

Researcher, Department of Surgery *May 2026 – Present*
Seoul National University Bundang Hospital Supervisor: Prof. So Hyun Kang

- Developing AI models for predicting Enhanced Recovery After Surgery (ERAS) outcomes using perioperative clinical data
- Investigating structured preoperative variables and longitudinal perioperative data for dynamic outcome prediction
- Exploring multimodal fusion of structured, longitudinal, and unstructured clinical data for integrated recovery prediction
- Applying explainable AI methods to identify clinically relevant factors associated with postoperative recovery

Research Intern, Center for Artificial Intelligence in Healthcare (CAIH) *Sep 2025 – Apr 2026*
Seoul National University Bundang Hospital Supervisor: Prof. Sejoog Kim

- Developed an LLM-based hybrid model for gastric cancer EMR processing
- Designed a structured extraction pipeline for pathology registry variables

Undergraduate Research Intern, PEM Lab, Hongik University *Feb 2024 – Feb 2025*

- Studied ECG characteristics and cardiac domain knowledge for arrhythmia classification using the MIT-BIH Arrhythmia Database
- Contributed to ECG signal preprocessing and deep learning model development for arrhythmia detection

Training

Biohealth AI Academy, AWS × Sungkyunkwan University SAY *Feb 2025 – Sep 2025*

- Participated in AWS official training on cloud computing and AI-based data analysis
- Conducted pre-project and final project using AWS cloud services for biohealth AI

Publications & Conference Abstracts

Structured LLM Pipeline for Automated Gastric Cancer Registry Construction from Bilingual Pathology Reports: A Pilot Study

Hungi Jung, So Hyun Kang, Soon jae Kwon, Mira Yoo, Dongjae Jeon, Young suk Park, Tae-han Kim, Yun-suhk Suh

KINGCA WEEK 2026 Abstract Track (Accepted)

Structured LLM Pipeline for Automated Gastric Cancer Registry Construction from

Bilingual Pathology Reports

Hungi Jung, So Hyun Kang, Myeong ju Kim
MLHC 2026 Clinical Abstract Track (Accepted)

Wavelet-Based ECG Feature Extraction and CNN for Arrhythmia Classification: An Enhanced Approach Using MIT-BIH Database

YunSeo Jo, Hungi Jung, SeungJu Oh, HaYoon Song
Annual Conference of KIPS, 2024 [Link]

Projects

Extraction and Structuring of EMR Data Using a Query–Regex–LLM Hybrid Model *Sep 2025 – Apr 2026*

- Extracted structured clinical data from SNUBH CDW to build registry-ready datasets
- Designed a hybrid pipeline combining regular expressions and LLMs for EMR parsing
- Implemented data transformation logic to standardize heterogeneous clinical records
- Developed a scalable pipeline for automated registry data generation

ML-based Mortality Prediction Dashboard using MIMIC-IV *Jul 2025 – Sep 2025*

- Developed mortality prediction models using longitudinal clinical data from MIMIC-IV
- Trained XGBoost, Transformer, and TCN models; selected optimal model by performance metrics
- Applied SHAP and LIME for model explainability
- Developed an AI-based dashboard for healthcare professionals

CNN + XAI-based Brain Tumor Detection *Jun 2025 – Aug 2025*

- Utilized an open-access brain tumor dataset from Kaggle
- Designed preprocessing and classification tasks using CNN models
- Applied Grad-CAM for model interpretability

AI-based Arrhythmia Detection Using MIT-BIH Dataset *Jun 2024 – Apr 2025*

- Designed a deep learning model for arrhythmia detection from ECG data
- Built a preprocessing pipeline incorporating cardiac domain knowledge and signal characteristics

Research Interests

Multimodal AI for Healthcare · Clinical Machine Learning · Longitudinal EHR & Time-series Modeling
Clinical NLP & Large Language Models · Explainable AI

Skills

Languages: Python, Java, SQL

ML/DL: PyTorch, Hugging Face, vLLM, scikit-learn, pandas, numpy, wfdb

Tools & Infra: Git, Docker, VS Code, Linux/Bash, CUDA, Weights & Biases (W&B)

Cloud: AWS